An Attention-Guided Neck Feature Fusion Framework for Drone Detection Using YOLOv8 and CBAM

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***Abstract*—Small unmanned aerial vehicles (UAVs) can be challenging to detect in outdoor images, particularly when they appear at a distance and occupy only a small portion of the frame. Their appearance can also be easily confused with birds, vegetation, clouds, buildings, and other background elements. To address this challenge, this paper proposes an attention-guided enhancement of the You Only Look Once (YOLO) detector YOLOv8x by incorporating the Convolutional Block Attention Module (CBAM) at two selected feature-fusion stages in the path aggregation neck. Specifically, channel attention followed by spatial attention is introduced after the 40×40 and 80×80 feature-fusion stages, allowing the network to refine the multi-scale features before they are passed to the detection head. Rather than making extensive changes to the YOLOv8x backbone, the proposed approach retains the original feature hierarchy and adds 64,198 parameters (0.094%) and 0.0018 GFLOPs. On the original validation split, the proposed model reaches a mean average precision of 92.93% at mAP@0.5 and 57.10% at mAP@0.5:0.95, 3.71 and 3.34 percentage points above the reported baseline. We further audit the dataset, find near-duplicate images shared by the original training and validation sets, and provide a leakage-free, scene-grouped train/validation/test split together with a fully scripted evaluation protocol. The ablation study shows that channel and spatial attention contribute in different but complementary ways.**

***Index Terms*—drone detection, UAV detection, YOLOv8, CBAM, attention mechanism, object detection, small object detection, feature fusion**

I. Introduction

Unmanned aerial vehicles (UAVs), or drones, are now used well beyond military settings, in tasks such as infrastructure inspection, precision agriculture, package delivery, mapping, disaster response, and surveillance. Their low cost and agility, however, also raise security concerns, including intrusion into restricted airspace, smuggling, covert surveillance, and collision risks near airports. This growth is part of a wider rise of small autonomous robots, which now extends even to bioinspired microrobots for medical use [1].

Spotting low-flying drones in cluttered outdoor scenes is still difficult. Seen from a ground camera, a drone may cover only 16×16 to 32×32 pixels of a high-definition frame, so fine details such as propellers, motor mounts, and the airframe are largely lost after downsampling. Its appearance also changes with viewing angle, lighting, motion blur, and partial occlusion by trees, poles, or buildings, while clouds, birds, and moving foliage can produce drone-like patterns that cause false alarms or missed targets.

Preserving information across scales is therefore essential. Feature Pyramid Networks (FPN) [2] and Path Aggregation Networks (PANet) [3] merge features from different resolutions, but this fusion can also carry background noise into the feature maps and blur the edges of small objects. Attention mechanisms address this by weighting features according to their channel and spatial importance, and the Convolutional Block Attention Module (CBAM) [4] does so by applying channel attention followed by spatial attention.

In this work, two CBAM modules are added to the path aggregation neck of YOLOv8x, after the 40×40 fusion stage at Layer 12 and the 80×80 fusion stage at Layer 15. These are the points where features from different scales meet, so refining them here is intended to suppress background clutter and keep the visual cues of drones before detection.

The main contributions of this work are as follows:

- We propose YOLOv8-CBAM, an attention-guided architecture for UAV detection in visually complex outdoor environments.
- We introduce a targeted neck-level attention strategy by placing two CBAM modules after Layers 12 and 15, enhancing the fused feature representations while keeping the original YOLOv8x architecture unchanged.
- We conduct an ablation study comparing the baseline YOLOv8x with channel-only attention, spatial-only attention, and the complete CBAM configuration, and provide scripts for placement, attention-type, multi-seed and state-of-the-art comparisons on a leakage-free split.
- We report measured complexity: CBAM adds 64,198 parameters (+0.094%) and 0.0018 GFLOPs (+0.0007%) to the 68.15 M-parameter, 258.11-GFLOP detector, and we audit the benchmark for train/validation near-duplicates.

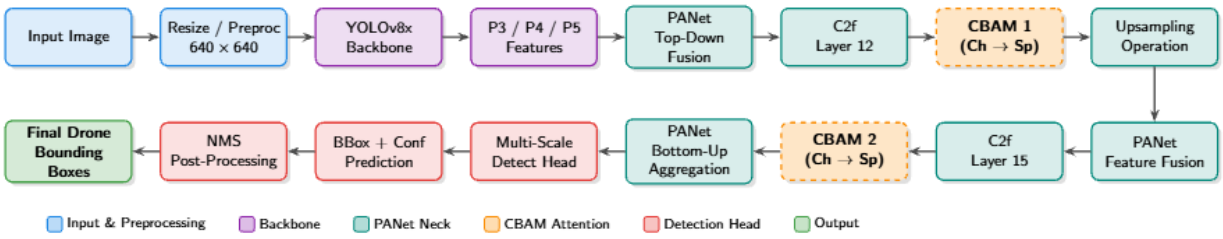


Fig. 1: End-to-end pipeline of the proposed YOLOv8-CBAM drone detection framework.

II. Related Work

A. Deep Learning Object Detection

Object detectors built on deep learning are usually grouped into two-stage and single-stage designs. Two-stage methods such as Faster R-CNN [5] first propose candidate regions and then classify and refine them, which gives good accuracy but slows down inference. Single-stage methods such as SSD [6] and the YOLO family predict classes and boxes in a single pass, which makes them better suited to real-time use. YOLOv4 [7] brought in a CSPNet backbone, PANet feature routing, and mosaic augmentation, while YOLOv7 [8] improved efficiency through extended efficient layer aggregation. YOLOv8 [9] then added the C2f block for better gradient flow and feature reuse, together with an anchor-free decoupled head; YOLO11 and the transformer-based RT-DETR [16] are used here as additional comparison detectors.

B. Multi-Scale Representation and Attention

Small objects are hard to detect because deeper layers gain semantic information at the cost of spatial resolution. FPN [2] handles this with a top-down path and lateral connections that merge features across scales, and PANet [3] adds a bottom-up path so that low-level spatial detail reaches the deeper layers more easily. Improved YOLOv5 variants have applied similar ideas to general object detection [10] and to finding small defects on cluttered printed circuit boards [11].

Attention modules take a different route by weighting features according to how useful they are. Squeeze-and-Excitation (SE) networks [12] learn channel weights from globally pooled features, while CBAM [4] adds a spatial branch so that the network learns both which channels and which locations matter. Coordinate Attention [13] builds position information into channel attention by pooling along each spatial direction, and CBAM SAUNet [14] showed that combining channel and spatial attention helps with difficult corner cases in segmentation.

III. Dataset and Problem Formulation

A. Dataset Attributes and Annotation

This study uses the publicly available YOLO Drone Detection Dataset from Kaggle [15], which contains 1,359 images of a single class, drone, captured against clear and overcast skies, urban skylines, and dense vegetation. The original release is split into 1,012 training (74.5%) and 347 validation (25.5%) images with 1,117 and 369 boxes, respectively. Each drone is labeled with a normalized bounding box

$$b = (x_c, y_c, w, h), \tag{1}$$

where (x_c, y_c) is the box center and w and h are its width and height. All images are letterbox-resized to 640×640 pixels so that their aspect ratio is preserved.

As Fig. 2 shows, box centers cluster near the middle of the frame and box sizes vary widely. Measured after letterboxing to 640 px with the COCO thresholds, only 13 of the 1,117 training boxes (1.2%) and 25 of the 369 validation boxes (6.8%) are small ($< 32^2$ pixels); 81.8% of the validation boxes are large ($\geq 96^2$). The benchmark is therefore dominated by close-range drones, which limits how strongly it can test small-object detection (Section VIII-A).

Provenance and near-duplicate audit. The images come from several sources whose file names identify them: web photographs (`pic_*`, numeric names) and frames sampled from videos (`foto*`, `yoto*`, `scene*`). We hashed every image with a 64-bit perceptual hash (pHash) and compared all train–validation pairs (Table IV). In the original split, 33 of the 347 validation images (9.5%) have a near-duplicate (Hamming distance ≤ 4) in the training set, 30 of them identical pictures stored under different names, and 48 (13.8%) have a same-scene match (≤ 10). We therefore built a *scene-grouped* split: images linked by a pHash distance ≤ 10 or by neighbouring frame numbers (< 100 apart) of the same video source are merged into groups, and whole groups are assigned, per source family, to train/validation/test in a 70/15/15 ratio. The result (975/197/187 images) has no cross-split pair within 10 bits. Model selection uses only the validation part; the test part is evaluated once.

B. Formal Detection Objective

The detector is trained by minimizing a combined loss function:

$$L_{total} = \lambda_{box} L_{CIoU} + \lambda_{cls} L_{BCE} + \lambda_{dfl} L_{DFL}. \tag{2}$$

Here, Complete Intersection over Union (CIoU) loss is used for bounding-box localization, Binary Cross-Entropy (BCE) loss handles classification, and Distribution Focal Loss (DFL) helps improve localization; the Ultralytics defaults $\lambda_{box} = 7.5$, $\lambda_{cls} = 0.5$, $\lambda_{dfl} = 1.5$ are used.

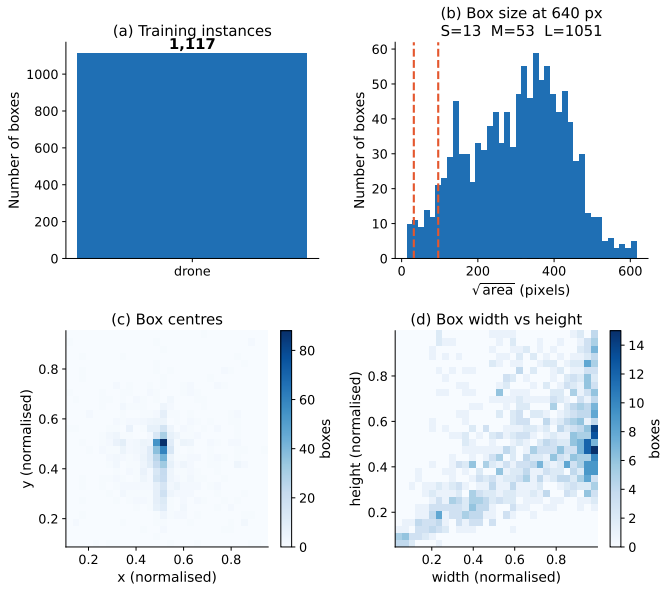


Fig. 2: Statistics of the training annotations: (a) number of drone instances, (b) box size after letter-boxing to 640 px with the COCO small/large borders (dashed), (c) distribution of normalized box centers, and (d) joint distribution of normalized box width and height.

IV. Proposed YOLOv8-CBAM Framework

A. Overall Architectural Pipeline

As shown in Fig. 1, YOLOv8-CBAM keeps the standard YOLOv8x backbone and decoupled anchor-free detection head, and changes only the path aggregation neck. Two CBAM modules are placed after Layer 12 and Layer 15, where the 40×40 and 80×80 features are fused, so that the combined features are refined before they pass through the bottom-up pathway to the detection head and non-maximum suppression.

B. Backbone and Multi-Scale Feature Hierarchy

YOLOv8x is used as the base network. Its backbone is built using modified CSPNet blocks called C2f. As the input passes through the backbone, it is progressively downsampled by factors of 8, 16, and 32, producing three feature pyramid stages (channel counts from a tensor-shape trace, Table I):

- P3 (80×80, 320 channels): high-resolution features with fine spatial details.
- P4 (40×40, 640 channels): medium-resolution features that balance spatial information with semantic context.
- P5 (20×20, 640 channels): lower-resolution, deeper features with wider receptive fields through the Spatial Pyramid Pooling-Fast (SPPF) module.

C. CBAM Formulation

The internal operation of the CBAM module is illustrated in Fig. 3. For an intermediate feature tensor $F \in \mathbb{R}^{C \times H \times W}$, CBAM applies channel attention followed by spatial attention.

TABLE I: Principal feature stages of YOLOv8x for a 640×640 input (measured by a tensor-shape trace).

Layer	Module	Channels	Resolution	Role
0	Conv (k=3, s=2)	80	320×320	Low-level stem
1	Conv (k=3, s=2)	160	160×160	Downsampling
2	C2f (n=3)	160	160×160	Feature extraction
3	Conv (k=3, s=2)	320	80×80	Downsampling
4	C2f (n=6)	320	80×80	P3 multi-scale feature
5	Conv (k=3, s=2)	640	40×40	Downsampling
6	C2f (n=6)	640	40×40	P4 multi-scale feature
7	Conv (k=3, s=2)	640	20×20	Downsampling
8	C2f (n=3)	640	20×20	Deep context extraction
9	SPPF (k=5)	640	20×20	P5 multi-scale feature
12	C2f	640	40×40	Neck fusion / CBAM-1
15	C2f	320	80×80	Neck fusion / CBAM-2
22	Detect head	Multi	Multi	Decoupled detection

The channel attention map $M_c \in \mathbb{R}^{C \times 1 \times 1}$ is obtained by applying average and max pooling across the spatial dimensions. These pooled features are passed through a shared multilayer perceptron (MLP) with a reduction ratio of $r = 16$, a ReLU activation δ between its two layers, and no bias terms, and the results are combined and passed through a sigmoid function σ:

$$M_c(F) = \sigma\Big(W_1\,\delta\big(W_0(F^c_{avg})\big) + W_1\,\delta\big(W_0(F^c_{max})\big)\Big), \quad (3)$$

where $W_0 \in \mathbb{R}^{C/r \times C}$ and $W_1 \in \mathbb{R}^{C \times C/r}$ are the shared MLP weights. The resulting channel attention is applied to the input feature map through element-wise multiplication:

$$F' = M_c(F) \otimes F. \quad (4)$$

Next, the spatial attention module applies average and max pooling along the channel dimension, concatenates the two maps, and processes them using a 7×7 convolution followed by a sigmoid function:

$$M_s(F') = \sigma\big(f^{7\times7}([\mathrm{AvgPool}_c(F'); \mathrm{MaxPool}_c(F')])\big). \quad (5)$$

The spatial attention map is applied to the refined feature map:

$$F'' = M_s(F') \otimes F'. \quad (6)$$

Finally, the CBAM output is blended with its input through a learnable scalar gate γ that is initialized to zero,

$$Y = F + \gamma\,(F'' - F), \quad (7)$$

so that at the start of fine-tuning the network reproduces the pretrained YOLOv8x exactly and learns how much attention to apply.

D. Targeted Attention Placement

Instead of adding attention modules throughout the network, we place two CBAM modules at feature-fusion points in the neck:

- CBAM-1 (Layer 12): placed after the C2f Layer 12 and operating on a $640 \times 40 \times 40$ feature map, where upsampled P5 features are combined with P4 features.
- CBAM-2 (Layer 15): placed after the C2f Layer 15 and operating on a $320 \times 80 \times 80$ feature map, where top-down

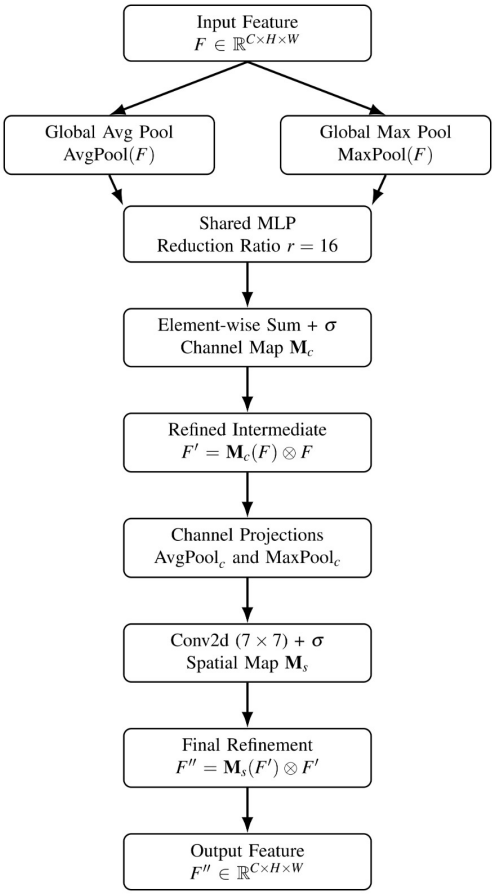


Fig. 3: CBAM sequential channel-then-spatial attention workflow.

semantic information is combined with high-resolution P3 features before the bottom-up aggregation stage.

E. Attention Integration and Training

The attention modules are attached by *module wrapping*: the C2f modules at indices 12 and 15 of the Ultralytics layer list are replaced by a wrapper that first runs the original C2f block and then the gated CBAM of Eq. (7). The wrapper copies the layer's routing attributes, so the rest of the network is unchanged, and the same code switches between full CBAM, channel-only, spatial-only and baseline execution. The wrapper is inserted *after* the COCO-pretrained YOLOv8x weights are loaded, which preserves the pretrained initialization. The original weights are *not* frozen: all parameters (except the fixed DFL projection) are fine-tuned jointly with CBAM. We verified that (i) all CBAM tensors, including both gates γ, are registered in the optimizer, (ii) the wrapped modules and gate values are stored in and restored from Ultralytics checkpoints, and (iii) the model exports to ONNX (opset 17) with a maximum deviation of 9.7×10^{-4} from PyTorch. Because stock Ultralytics rebuilds the network from its YAML file at the start of training, the wrapping is applied inside a custom trainer's `get_model()`.

TABLE II: Hyperparameter and Training Specifications

Hyperparameter	Setting
Base model	YOLOv8x (COCO-pretrained)
Input size	640×640 (letterbox)
Training epochs	32
Batch size	16
Optimizer	SGD, momentum 0.937*
Weight decay	0.0005
Initial learning rate η_0	0.01*
Channel reduction ratio r	16
Spatial kernel	7×7
Classes	1 (drone)
Checkpoint	best fitness on validation
Seed	0 (0, 1, 2 in Table VI)

*The M1 baseline of Table III was trained with Ultralytics 8.0.141 and `optimizer=auto`, which selected AdamW ($\eta_0 = 0.002$, momentum 0.9).

V. Implementation and Training Setup

A. Training Protocol and Hyperparameters

The experiments were conducted using PyTorch with automatic mixed precision, with the YOLOv8x network initialized from COCO-pretrained weights. The settings are summarized in Table II. The Ultralytics default augmentation was used: mosaic (probability 1.0, disabled for the last 10 epochs), horizontal flip (0.5), random scaling ($\pm 50\%$), translation ($\pm 10\%$) and HSV perturbation (0.015, 0.7, 0.4), with a 3-epoch warm-up and a linear learning-rate decay to $0.01\eta_0$. For every configuration the checkpoint with the best Ultralytics fitness ($0.1\,\mathrm{mAP@0.5} + 0.9\,\mathrm{mAP@0.5{:}0.95}$) on the validation data is reported.

B. Evaluation Protocol and Metrics

The detection performance was evaluated using Precision (P), Recall (R), Average Precision (AP), mAP@0.5, and mAP@0.5:0.95. The AP is the area under the precision–recall curve; mAP@0.5 uses an IoU threshold of 0.5, and mAP@0.5:0.95 averages over IoU thresholds from 0.5 to 0.95. All metrics are computed with the Ultralytics validator using a confidence threshold of 0.001, class-aware NMS at IoU 0.7 and at most 300 detections per image. P and R are reported at the confidence that maximizes F1. The confusion matrix is drawn at a confidence of 0.25 with an IoU match threshold of 0.45, so its P/R differ from the tabulated max-F1 values. The baseline M1 used Ultralytics 8.0.141; the scripted revision experiments use Ultralytics 8.4.201 with PyTorch 2.15. GFLOPs are reported as $2\times$ multiply–accumulate operations, profiled with thop directly at 640×640; AP by object size uses pycocotools with the COCO area thresholds applied to boxes rescaled to the 640 px frame.

VI. Experimental Results and Analysis

A. Main Benchmark Comparison

Table III compares the baseline YOLOv8x (M1) with the proposed YOLOv8-CBAM (M4) on the original validation split; its last row lists the gain over the baseline. The proposed model improves all four metrics, with Precision rising by 4.48 and Recall by 4.85 percentage points (pp). Because M1

was trained in a separate run with a different optimizer and because of the near-duplicates found in the original split, these numbers should be read together with the matched, leakage-free comparison in Tables V–VII.

B. Ablation Study on Attention Components

To separate the effect of each attention branch, we compare four settings: the plain baseline (M1), channel attention only (M2), spatial attention only (M3), and full CBAM (M4), each applied after Layers 12 and 15. Table III shows that both single branches improve on the baseline, with the spatial branch gaining a little more than the channel branch, while the full CBAM gives the best results on every metric.

Placement, attention type and repeatability. Table V compares CBAM at Layer 12 only, at Layer 15 only and at both, as well as SE and Coordinate Attention at Layers 12+15, under identical settings on the scene-grouped split. Table VI reports the baseline and CBAM over three seeds, and Table VII compares YOLOv8x-CBAM with YOLOv5x, YOLO11x and RT-DETR-X trained under the same protocol.

C. Computational Complexity Analysis

The two CBAM modules add exactly 64,198 parameters: 51,299 at Layer 12 ($2 \cdot 640 \cdot 40$ MLP weights, 98 convolution weights and one gate) and 12,899 at Layer 15, raising the 1-class YOLOv8x from 68,153,571 to 68,217,769 parameters (+0.094%). Profiled directly at 640×640, the cost rises from 258.1123 to 258.1141 GFLOPs (+0.0018 GFLOPs, +0.0007%); fvcore gives the same increment. The element-wise operations of the attention (pooling, sigmoid gating and blending), which these profilers do not count, add a further 0.028 GFLOPs. The profile reported by Ultralytics' `model.info()` (+0.10 GFLOPs) is an artifact of extrapolating from a 32×32 proxy image: the cost of the channel MLP does not grow with image size but is multiplied by $(640/32)^2$.

A small arithmetic overhead does not guarantee a negligible runtime overhead, so we also timed the network (Table VIII). On a 4-thread CPU (FP32), the median latency is 604.59 ms for the baseline and 606.91 ms for YOLOv8-CBAM; the median paired difference, +7.67 ms (+1.27%), is well below the run-to-run standard deviation of about 50–58 ms.

VII. Visual and Qualitative Analysis

A. Training Convergence Dynamics

Fig. 4 illustrates the training convergence of the baseline over 32 epochs. The loss curves decrease steadily, and precision, recall and mAP@0.5:0.95 increase over training; validation mAP@0.5:0.95 is still rising at the last epoch (0.528 to 0.538), so a longer schedule may further improve the models.

B. Confusion Matrix and PR Curves

Fig. 5 shows the confusion matrix and the precision–recall curve of the *baseline* on the original validation split. At the confusion-matrix operating point (confidence 0.25, IoU match 0.45) the model finds 320 of the 369 drones and produces 48 false positives, i.e. P = 86.96% and R = 86.72%; these differ from Table III, which reports P/R at the max-F1 confidence. Because the validation set contains no drone-free images, the 48 false positives are spurious boxes on images that do contain a drone; the matrix does not measure rejection of empty scenes.

C. Qualitative Detection Performance

Fig. 6(a) shows a representative detection produced by YOLOv8-CBAM for a close-range quadcopter captured against a clear sky, with out-of-focus palm foliage in the background. The model detects the drone with a confidence score of 0.79, and the textured vegetation behind the airframe does not trigger additional detections. The predicted box covers the main airframe and all four rotors but excludes part of the lower gimbal camera and the landing skids, showing that box boundaries can be imprecise for large near-field targets even when the detection is correct, consistent with the gap between mAP@0.5 and mAP@0.5:0.95.

D. Detection in Cluttered Ground-Level Scenes

Fig. 6(b) shows a ground-level scene in which a multirotor drone with extended landing gear hovers at medium range above a dense tree line, with several people and pieces of equipment in the foreground. Despite the low evening illumination, the model localizes the drone with a tight bounding box, and no detections are produced on the people, trees, or ground equipment. The confidence of 0.62 is lower than in Fig. 6(a), as the drone occupies a smaller portion of the frame and its thin arms and legs provide fewer distinctive cues.

Failure cases. Missed or poorly localized drones occur mainly under severe motion blur, at extreme distances where the drone occupies fewer than 8×8 pixels, and under heavy occlusion by building structures. Their frequency has not yet been quantified; the benchmark contains few such cases (Section VIII-A).

VIII. Discussion

Placing attention only at the neck fusion points is motivated by the observation that plain concatenation of multi-scale features can carry background into the higher-resolution maps; refining the fused features after Layers 12 and 15 is intended to suppress this clutter while keeping fine spatial detail. The ablation suggests that spatial localization matters slightly more than channel selection for compact drone silhouettes, though both work best together.

A. Limitations

The study uses a single, small public dataset. Although the scene-grouped split removes the near-duplicates we found, all splits come from the same sources, so generalization to other cameras, sensors (e.g. infrared) and environments is not established; evaluation on additional drone datasets is future work. The benchmark contains few small targets (6.8% of validation boxes below 32^2 pixels), so it cannot show a small-object benefit conclusively, and it contains no

TABLE III: Ablation of attention configurations on the original validation split (347 images). Params and GFLOPs are measured for the 1-class model (640×640, GFLOPs $= 2\times$MACs, thop); accuracy values are those of the submitted version. Gains are in percentage points (pp).

Configuration	Params	GFLOPs	P (%)	R (%)	mAP@0.5 (%)	mAP@0.5:0.95 (%)
M1: Baseline[†]	68,153,571	258.1123	86.82	83.90	89.22	53.76
M2: Channel-only	68,217,573	258.1126	88.54	85.12	90.71	54.92
M3: Spatial-only	68,153,769	258.1139	89.15	86.40	91.25	55.43
M4: YOLOv8x-CBAM	68,217,769	258.1141	91.30	88.75	92.93	57.10
Gain (M4−M1)	+64,198	+0.0018	+4.48 pp	+4.85 pp	+3.71 pp	+3.34 pp

[†]M1 is the final epoch of a separate run made with Ultralytics 8.0.141 and `optimizer=auto` (AdamW, $\eta_0 = 0.002$), whereas the ablation script used for M2–M4 specifies SGD ($\eta_0 = 0.01$); no raw outputs of M2–M4 were retained. The matched-protocol comparison on a leakage-free split is Table VI.

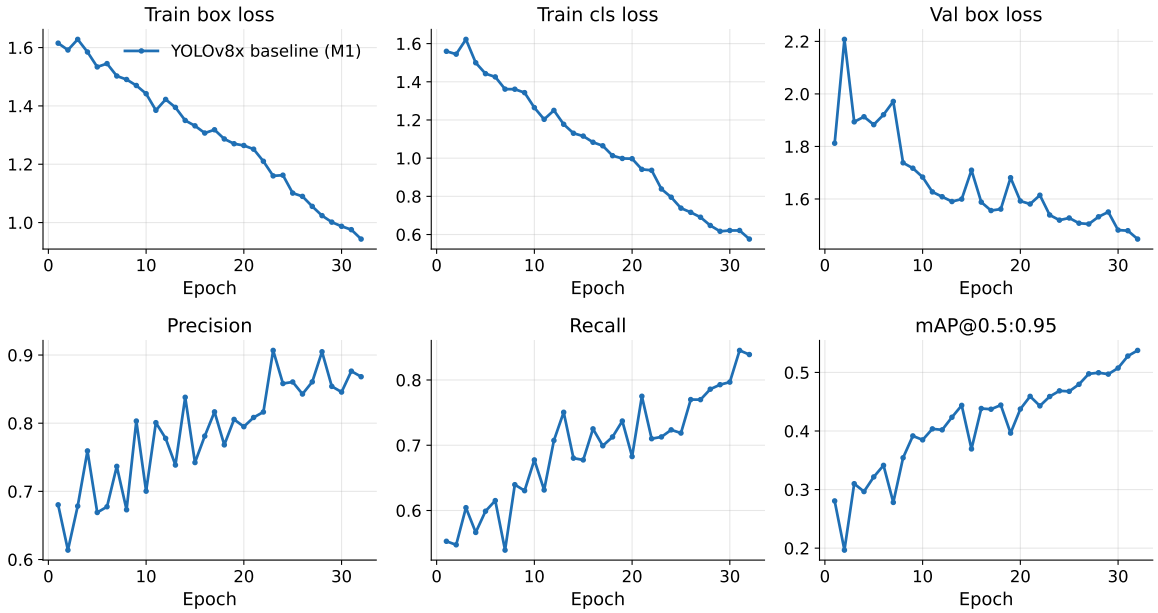


Fig. 4: Training losses and validation metrics of the baseline YOLOv8x (M1) over its 32 training epochs.

TABLE IV: Near-duplicate audit of the original split (pHash, Hamming ≤ 4 / ≤ 10 bits) and the new scene-grouped split used for all retraining.

Original split (1,012 train / 347 val)			
Val images with near-duplicate in train	33 (9.5%)		
Val images with same-scene match in train	48 (13.8%)		
Scene-grouped split	Train	Val	Test
Images	975	197	187
Boxes	1042	235	209
Cross-split pairs ≤ 10 bits		0	

TABLE V: Placement and attention-type ablation on the test split (seed 0). Added parameters are measured. Cells marked "–" require trained checkpoints / a GPU and were not run in this revision.

Configuration	Added params	mAP@0.5	mAP@0.5:0.95
None (baseline)	0	–	–
CBAM @ L12	–	–	–
CBAM @ L15	–	–	–
CBAM @ L12+L15	+64,198	–	–
SE @ L12+L15	+65,022	–	–
CoordAtt @ L12+L15	+50,012	–	–

TABLE VI: Test-split results over three seeds (mean ± sample std, %). Cells marked "–" require trained checkpoints / a GPU and were not run in this revision.

Model	P	R	mAP@0.5	mAP@0.5:0.95
YOLOv8x	–	–	–	–
YOLOv8x-CBAM	–	–	–	–

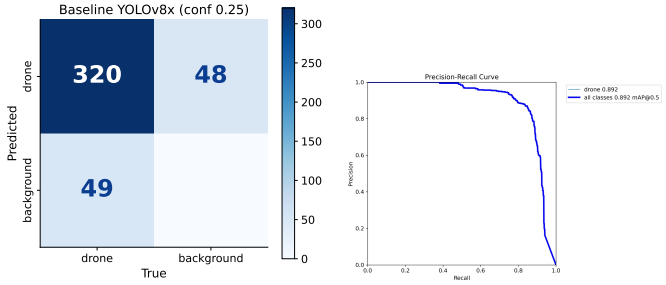
drone-free images and few bird-like distractors, so false-alarm behavior on empty skies, birds and aircraft is not measured. We therefore do not claim improved background rejection; a dedicated negative-image and distractor test set is needed for that. Finally, the matched multi-seed, placement and state-of-the-art experiments were scripted but must be run on a GPU to complete Tables V–IX.

TABLE VII: Comparison with state-of-the-art detectors on the held-out test split of the scene-grouped split (same epochs, input size, batch size and augmentation; seed 0). Cells marked "–" require trained checkpoints / a GPU and were not run in this revision.

Model	Params (M)	GFLOPs	FPS	P (%)	R (%)	mAP@0.5 (%)	mAP@0.5:0.95 (%)
YOLOv5x (u)	–	–	–	–	–	–	–
YOLO11x	–	–	–	–	–	–	–
RT-DETR-X	–	–	–	–	–	–	–
YOLOv8x	–	–	–	–	–	–	–
YOLOv8x-CBAM (ours)	–	–	–	–	–	–	–

TABLE VIII: Measured complexity and runtime (batch 1, 640×640, network forward only, Conv+BN fused, median of interleaved runs).

Model	Device	Prec.	ms/img	FPS	Peak mem (MB)
Baseline	CPU, 4 thr.	FP32	604.59	1.65	n/a
CBAM	CPU, 4 thr.	FP32	606.91	1.65	n/a
Params added by CBAM: +64,198 (+0.094%)					
GFLOPs added by CBAM: +0.0018 (+0.0007%)					



(a) Confusion matrix (b) Precision–Recall curve

Fig. 5: Validation results of the baseline YOLOv8x (M1): (a) confusion matrix at confidence 0.25 and (b) precision–recall curve (mAP@0.5 = 0.892).

TABLE IX: COCO-style AP@0.5:0.95 and AR by box size, measured after letter-boxing to 640 px (S: $< 32^2$, M: 32^2–96^2, L: $\geq 96^2$ pixels). Cells marked "–" require trained checkpoints / a GPU and were not run in this revision.

Model	AP_S	AP_M	AP_L	AR_S	AR_M	AR_L
Val boxes (count)	25	42	302			
Baseline	–	–	–	–	–	–
YOLOv8x-CBAM	–	–	–	–	–	–

IX. Conclusion

This paper presented YOLOv8-CBAM, which adds CBAM after the 40×40 and 80×80 neck fusion stages of YOLOv8x while leaving the backbone unchanged. FPN and PANet-style necks fuse multi-scale features without filtering them, channel-only schemes such as SE networks reweight features without asking where the target is, and attention is often inserted throughout the backbone. In contrast, our approach applies both channel and spatial attention only at the two points where multi-scale features merge, and attaches it through module wrapping with a zero- initialized gate, starting from the pretrained weights and fine-tuning the whole network. On the original validation split, the model reaches 92.93% mAP@0.5 and 57.10% mAP@0.5:0.95, gains of 3.71 and 3.34 percentage points over the reported baseline, for 64,198 extra parameters (+0.094%) and 0.0018 extra GFLOPs. We also release a leakage-free split and scripts that measure parameters, GFLOPs, latency, size-specific AP, seed variability and state-of-the-art comparisons, so that the gains can be verified under matched conditions.

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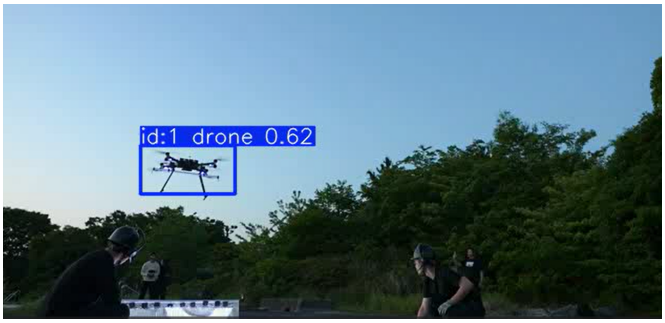
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(a) Close-range quadcopter against a clear sky



(b) Multirotor drone above a tree line

Fig. 6: Detection results of YOLOv8-CBAM. The frames were taken from a video-tracking demonstration; the leading number in each label is the tracker ID and is not part of the detection method.

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